# Yasith Silva

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## OBJECTIVE

I am a dedicated electronic engineering student, passionate in **Embedded Systems Design, Circuits Design, IoT Systems and Digital Systems Design**. As a team player with good leadership qualities, communication and presentation skills, and experience, I can adapt and learn quickly. I seek opportunities to contribute to the ever-evolving and exciting world of embedded systems, learning along the way.

## TECHNICAL SKILLS

**Skills:** Circuit Design - Schematics and PCBs, Embedded Systems Design, Programming, Firmware Development (With Mistra Standards), Digital Systems Design, FPGA Implementation

**Circuit Design**: Altium Designer, KiCAD, High speed design, Multilayer Design, Hierachical Schematic Design
**Embedded Systems**: Embedded C, Baremetal, FreeRTOS
**Programming Languages**: C/C++, Python, JavaScript
**Other**: GitHub, Experience using Linux (Ubuntu, Raspberry Pi OS)

## EXPERIENCE

**Trainee Associate Electronic Engineer** — *Zone24x7 (pvt) ltd*
Dec 2024 - Jul 2025
- Worked on IoT, UWB localisation and Embedded System
- Coded firmware for multiple projects following industry standards
- Got hands on experience with complex embedded systems
- Developed presentation skills through student visits, client demonstrations

**Course Instructor - Embedded Product Design for IoT** — *Skillsurf*
Dec 2024 - May 2025
- Did sessions on schematic and PCB design

**Freelance PCB Designer**
May 2024 - Present
- Designed and Developed Custom PCB Layouts for projects using *Altium Designer*
- Communicated effectively with clients to understand project requirements

**Visiting Instructor** — *University of Moratuwa*
Dec 2024 - May 2025
- Assisting telecommunication laboratory practices
- Managing the instructors for telecom practicals

**Co-founder** — *Metronix*
Jan 2025 - Present
- Looking into UI/UX design and marketing

## EDUCATION

**University of Moratuwa**, Sri Lanka
*BSc. (Hons) Engineering*, Electronic and Telecommunication — CGPA: 3.67/4.00
May 2022 - Present

**D.S. Senanayake College**, Colombo 07, Sri Lanka
G.C.E. Advanced Level (Physics, Mathematics, Chemistry) 3As / z-score 2.21
2012 - 2020

**Maris Stella College**, Bolawalana - Negombo, Sri Lanka
2007 - 2012

## COURSES AND CERTIFICATES

**Introduction to Embedded Machine Learning** — Edge Impulse
**FPGA Softcore Processors and IP Acquisition**
University of Colorado Boulder
**Hardware Description Languages for FPGA Designs**
University of Colorado Boulder
**Introduction to, Intermediate Machine Learning** — Kaggle
**MATLAB Onramp** — MathWorks

## PROJECTS

**Chip-Aware Universal Evaluation Platform (Ongoing)**:
FYP. A universal platform that can be reconfigured to evaluate a range of ICs, including motor controllers, ADCs, DACs, and the software required for configuring, and an AI agent to convert the datasheets to configuration files and assist testing.

- Designing the schematics, PCB and firmware for the expansion kit for the FPGAs, containing an ADC, a DAC, gain controllers, expanders and other inputs and outputs for the device under test.
- Designing the test cases

**METROBAND — A Metronome Wristband**:
An alternative to the traditional metronome which aids musicians to keep to their tempo, using vibrations. The circuit was designed around an ESP32-S2 chip, communicates with the phone using Bluetooth.

- Designed with *ESP32-S2-WROOM*
- Developed the firmware
- Schematics and PCB Design done using *Altium Designer*

**Wireless Reconfigurable Andon System with Maintenace Prediction**:
A System to flag problems in manufacturing lines. We equipped this with wireless data logging and maintenance prediction.

- Designed with *ESP32-S3-WROOM* chip
- Baremetal C coding with *ESP-IDF*
- Schematics and PCB Design using *Altium Designer*

**32-bit Single Precision Floating Point ALU**:
This project aims to create a 32-bit ALU for single precision floating point numbers, capable of addition, subtraction, multiplication and division.

- Addition, subtraction, multiplication in 3 cycles, division in 25
- Implemented on DE2-115

**32-bit Pipelined RISC-V Processor**:
A 32-bit soft-core processor on an FPGA to support RISC-V. Consists of the basic peripherals required to function.

- 5 Pipelined stages
- Supports RISC-V base instruction set
- Contains cache and branch prediction.

**UART Transceiver using DE0-Nano FPGA Board**:
UART Transceiver, communicating between two FPGA Boards. The design was modified to control a series of LEDs on each board from the other.

- Implemented on an *DE0-Nano* FPGA Board
- Design verified by a Simulating a Testbench on *Altera-ModelSim*

**EcoWatt - Energy Monitoring Gateway**:
A microcontroller-based energy monitoring gateway simulating a solar inverter interface, designed with constrained uplink, remote configuration, security, and FOTA support.
- Wrote firmware for remote configuration
- Kept abstraction layers and keeping in Mistra standards

**Flood-level Sensing System for Detection and Prediction:**
A system with several sensors to sense the water level of a body, and a rain gauge to predict the water level, and sends the collected data froma close area to a central server.
- Designed with *ESP12-E* chip
- Uses *MQTT* over Wifi to communicate with the server
- Schematics and PCB Design using *KiCad*

**Guitar Pedalboard**:
A set of fx pedals such as Overdrive, Fuzz, Tremolo and Wah. This was done as a fully analogue project, using operational amplifiers.
- PCB Design using *Altium Designer*
- Assembled and soldered the components

## COMPETITIONS

**VRCade 2.0 (Winners)**
***Signify*** — A text-to-sign language application for ease of communication for hearing impaired, designed for an AR glass, and a customized lightweight AR glass able to specifically run the AR application.
- Designed and customized the PCB for the AR Glass

**DVCon2025 India (Second Round)**
***A Custom Hardware Accelerator for U-Net*** — An accelerator for image segmentation using U-Net for Autonomous Driving Applications. This is to be integrated with the VEGA AT1051 - 32bit CPU IP core, running on a Genesys-2 board.
- Made the U-Net IP

**Idealize'24 (First Runners Up)**
**Spark Challenge 2023/24 (Finalists)**
***SportSense*** — A mobile app that helps users to keep correct poses and forms while doing their exercises using a body pose landmark identification model.
- Contributed in UI/UX design with *Android Studio*

**Brainstorm'24 (Finalists)**
***Project CrystalClear*** — A platform that consists of interactive exercises for Dyslexic patients using a finger-pose model evaluation.
- Contributed in UI design using *Codux*

**Uva Wellassa University Robot Battles 2.0 Death Race (Semifinalists)**
A Remote Controlled Battlebot on several races across an outdoor path containing various obstacles such as ramps, saws, hammers and fire.
- Designed the PCB and the wiring system
- Contributed in coding the RF communication part

## SOFT SKILLS

**Communication Skills**
- Language Skills
  - English — Bilingual proficiency
  - Sinhala — Native proficiency
- Presentation Skills
  - A previous member of the Zone24x7 Toastmasters Club

– Presented company projects at Zone24x7 at multiple student visits and in client demonstrations
– Was one of the presenters at the Mobitel Lab of the department during the EXMO-2023 exhibition.
– Presented Metroband—The metronome wristband at EXMO-2023
– Have been a presenter in several other project presentations and in competitions

**Leadership and Teamwork**

- In charge of the telecom lab instructors
- Was the leader of several group projects at the university.
- Was a batch coordinator for the Mobitel Lab at the EXMO-24 exhibition
- Was a part of the PR team of SLRC-2024
- Was one of the PR-IT Directors at the Rotaract Club of Achievers Lanka Business School for 2021/22
- Been on the organising committees at the school events.

## VOLUTEERING AND CLUBS

PR & IT Director — Rotaract Club of Acheivers Lanka Business School 2021/22
Video Editor — Electronics Club 2023
Member of Zone24x7 Toastmasters Club
Member of Association for Computer Machinery - University of Moratuwa
Member of IEEE Student branch - University of Moratuwa
Member of Classical Music Society - University of Moratuwa

**During School**

President — Catholic and Christian Society of D.S. Senanayake College 2019
Co-organizer — Aeronautical Society of D.S. Senanayake College 2019
Co-organizer — Western Music Society of D.S. Senanayake College 2019
Member of Senior Boys' Choir of D.S. Senanayake College

## REFERENCES

**Dr. Ajith Pasqual** *Ph.D. (Tokyo)*
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